

Auxiliary-predicted Compress Memory Model(**ApCM Model**): A Neural Memory Storage Model Based on Invertible Compression and Learnable Prediction

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Repository: <https://github.com/yauntyour/ApCM-py>

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Abstract

Current artificial intelligence systems, such as Large Language Models (LLMs), generally lack effective "runtime" memory mechanisms, making it difficult to adapt to dynamic and personalized interaction needs. To address this issue, this paper proposes **ApCM Model**—a novel neural memory storage architecture that integrates **Invertible Dimensionality Reduction** with a **Learnable Auxiliary Predictor**. An invertible neural network maps input data losslessly to a low-dimensional latent space, decomposing it into a **compressed representation** (z_{comp}) for storage and a discardable **auxiliary representation** (z_{aux}). The key innovation is the introduction of a lightweight predictor network that estimates z_{aux} from z_{comp} , enabling high-fidelity reconstruction of the original data via the inverse transform. Building on this, the work constructs a slot-based global memory bank (Memory Bank) and designs a cosine-similarity-based read mechanism along with an access-frequency-based write policy. Experiments show that ApCM Model, while maintaining compression efficiency comparable to Principal Component Analysis (PCA), exhibits stronger nonlinear modeling capabilities, offering a flexible, efficient, and learnable runtime memory solution for systems like LLMs.

Keywords: Runtime memory, Invertible Neural Networks, Flow models, Dimensionality compression, Memory bank, Large Language Models

1 Introduction

The rapid development of artificial intelligence, especially Large Language Models (LLMs), has made computational and storage resources key bottlenecks constraining further expansion of their capabilities. Current mainstream LLMs are essentially "stateless" systems, with their knowledge entirely solidified in the parameters obtained during training—i.e., "training-time memory." This paradigm shows limitations when dealing with tasks requiring long-context understanding, personalized interaction, or dynamic knowledge updates. A core characteristic of human intelligence is possessing powerful "runtime memory," enabling immediate storage, retrieval, and utilization of new information.

Therefore, endowing AI systems with similar memory capabilities and constructing efficient, learnable external memory modules has become a highly valuable research direction.

Traditional data compression methods like Principal Component Analysis (PCA) can effectively reduce dimensions, but their linear assumptions limit their ability to model complex data distributions. Furthermore, they typically involve lossy compression and lack the ability to optimize the reconstruction process through learnable mechanisms.

To address these challenges, this paper proposes the **ApCM Model** model. Its core idea is to **decouple memory storage from reconstruction and connect them via a learnable predictor**. Specifically, an invertible neural network based on coupling layers serves as the encoder, ensuring precise invertibility of the transformation. The input data, after encoding, is split into z_{comp} (the compressed storage part) and z_{aux} (the auxiliary information part). During storage, only z_{comp} is retained, and a lightweight network is trained to predict z_{aux} from z_{comp} . During reconstruction, the stored z_{comp} and the predicted z_{aux} are concatenated and passed through the inverse transform to recover the original data.

Furthermore, this paper constructs a global memory bank and designs a read mechanism based on cosine similarity along with a write policy based on access frequency, achieving dynamic memory management. The main contributions of this paper are as follows:

1. Proposes the ApCM Model architecture, integrating invertible compression with learnable prediction to realize an optimizable lossy-reconstruction memory paradigm;
2. Designs a complete memory read-write mechanism supporting content-based retrieval and frequency-based updates;
3. Validates through experiments that the model’s reconstruction performance on non-linear data surpasses traditional linear compression methods.

2 Model Architecture

ApCM Model aims to build a learnable, efficient, and flexible runtime memory system. Its architecture comprises two core components: the **Invertible Dimensionality Reduction and Predictor (IDRP)** and the **Memory Read-Write Controller**.

2.1 Invertible Dimensionality Reduction and Predictor (IDRP)

IDRP combines invertible transformations with a prediction mechanism to achieve efficient compression and high-fidelity reconstruction.

2.1.1 Invertible Network Encoder

The encoder consists of N stacked **affine coupling layers** and **random permutation layers**.

- **Affine Coupling Layer** (`InvertibleCouplingLayer`): Input $\mathbf{x} \in \mathbb{R}^d$ is split into $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^{d/2}$. A sub-network $\mathcal{N}$ generates scale factor $\mathbf{s}$ and translation factor $\mathbf{t}$ based on $\mathbf{x}_1$:

$$[\mathbf{s}, \mathbf{t}] = \mathcal{N}(\mathbf{x}_1) \quad (1)$$

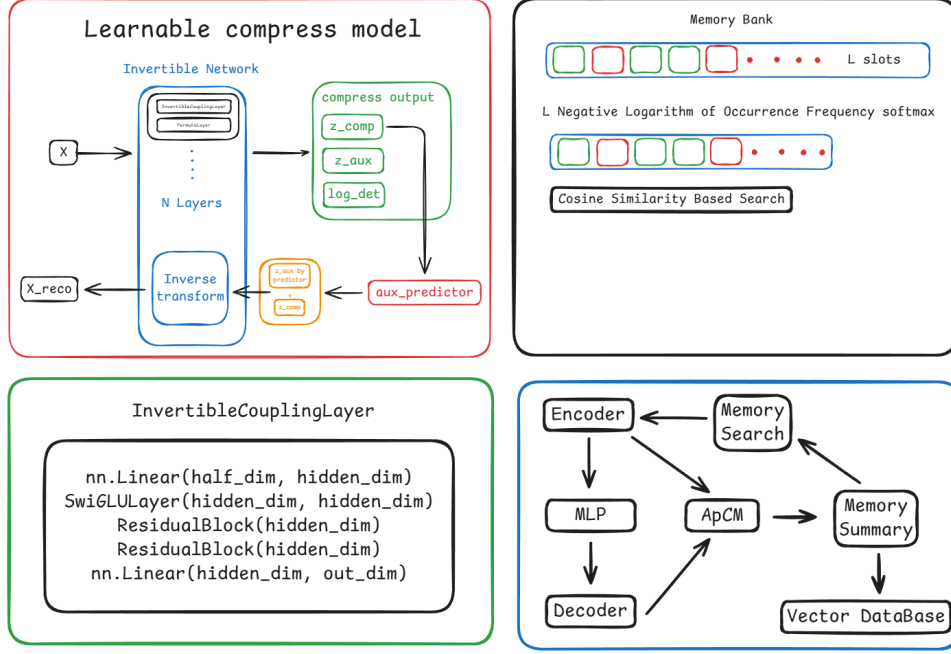


Figure 1: Architecture diagram of the Invertible Dimensionality Reduction and Predictor

The sub-network consists of linear layers, SwiGLU activation functions, and residual blocks. The transformation output is:

$$\mathbf{y}_1 = \mathbf{x}_1, \quad \mathbf{y}_2 = \mathbf{x}_2 \odot \exp(\mathbf{s}) + \mathbf{t} \quad (2)$$

This transformation is exactly invertible, with the inverse being $\mathbf{x}_2 = (\mathbf{y}_2 - \mathbf{t}) \odot \exp(-\mathbf{s})$.

- **Random Permutation Layer (PermuteLayer)**: Introduces a fixed random permutation after the coupling layer to ensure sufficient interaction between dimensions.

After N layers of transformation, the input $\mathbf{x}$ is mapped to a latent representation $\mathbf{z} = f_\theta(\text{flatten}(\mathbf{x}))$, where f_θ is the invertible network.

2.1.2 Latent Space Decomposition and Prediction

Split $\mathbf{z}$:

$$\mathbf{z} = [\mathbf{z}_{comp}, \mathbf{z}_{aux}] \quad (3)$$

$\mathbf{z}_{comp} \in \mathbb{R}^m$ is used for storage, and $\mathbf{z}_{aux}$ is discarded. This decomposition strategy is inspired by the gating mechanism in Mixture-of-Experts (MoE) architectures [1], where different parts of the representation serve distinct roles. Introduce an auxiliary information predictor g_ϕ (MLP structure: Linear $\rightarrow$ SwiGLU $\rightarrow$ Linear) to predict $\mathbf{z}_{aux}$ from $\mathbf{z}_{comp}$:

$$\hat{\mathbf{z}}_{aux} = g_\phi(\mathbf{z}_{comp}) \quad (4)$$

2.1.3 Workflow

- **Encoding**: $\mathbf{x} \xrightarrow{f_\theta} \mathbf{z} \rightarrow (\mathbf{z}_{comp}, \mathbf{z}_{aux_true})$

- **Compression:** Output $\mathbf{z}_{comp}$
- **Reconstruction:** $\mathbf{z}_{comp} \xrightarrow{g_\phi} \hat{\mathbf{z}}_{aux} \rightarrow \hat{\mathbf{z}} = [\mathbf{z}_{comp}, \hat{\mathbf{z}}_{aux}] \xrightarrow{f_\theta^{-1}} \hat{\mathbf{x}}$

By jointly optimizing f_θ and g_ϕ (e.g., minimizing reconstruction loss), the model learns to make $\mathbf{z}_{comp}$ contain sufficient information to accurately predict $\mathbf{z}_{aux}$.

2.2 Memory Read-Write Controller

Based on IDRP, a global memory bank $\mathcal{M} \in \mathbb{R}^{\max_mem \times m}$ is constructed, with each row storing a $\mathbf{z}_{comp}$.

2.2.1 Read Mechanism

- Encode the query $\mathbf{x}$ into $\mathbf{q} = \mathbf{z}_{comp}$
- Compute the cosine similarity between $\mathbf{q}$ and each vector in $\mathcal{M}$:

$$\text{sim}_i = \frac{\mathbf{q}^\top \mathcal{M}_i}{\|\mathbf{q}\|_2 \|\mathcal{M}_i\|_2} \quad (5)$$

- Take the slot $\mathcal{M}_{i^*}$ with the highest similarity, reconstruct $\hat{\mathbf{x}}_{mem}$ via IDRP, return it, and update its access frequency.

2.2.2 Write Mechanism

- Encode batch inputs $\{\mathbf{x}_j\}$ to obtain $\{\mathbf{z}_{comp}^{(j)}\}$, compute the mean $\bar{\mathbf{z}}$ as the write vector.
- Employ a "first idle, then least frequently used" strategy to select a slot:
 1. Prioritize selecting unused slots (`AFF_ctrl[i] == 0`)
 2. Otherwise, overwrite the slot with the lowest access frequency.
- Write $\bar{\mathbf{z}}$ into the selected slot and reset its access count.

3 Operational Mechanism and Principles

ApCM Model simulates the encoding, storage, retrieval, and reconstruction processes of human brain memory:

1. **Encoding and Separation:** The invertible network acts as a nonlinear encoder, reorganizing the input into a latent representation $\mathbf{z}$. By splitting $\mathbf{z}$, the model is forced to learn a representation where $\mathbf{z}_{comp}$ contains key information sufficient to infer $\mathbf{z}_{aux}$.
2. **Lossy Storage and Intelligent Reconstruction:** Only storing $\mathbf{z}_{comp}$ achieves compression. Reconstruction quality depends on predictor performance, optimized through end-to-end training, enabling lossy reconstruction that surpasses traditional linear methods.

3. **Memory Interaction:** Reading enables content-based associative retrieval; writing follows the "use it or lose it" principle, maintaining dynamic updates and effective utilization of the memory bank.

4 Experiments and Data Analysis

4.1 Synthetic Data Training (PCA for fitting), Real Data Testing

4.1.1 IDRП (Predictor pre-trained for 2000 epochs):

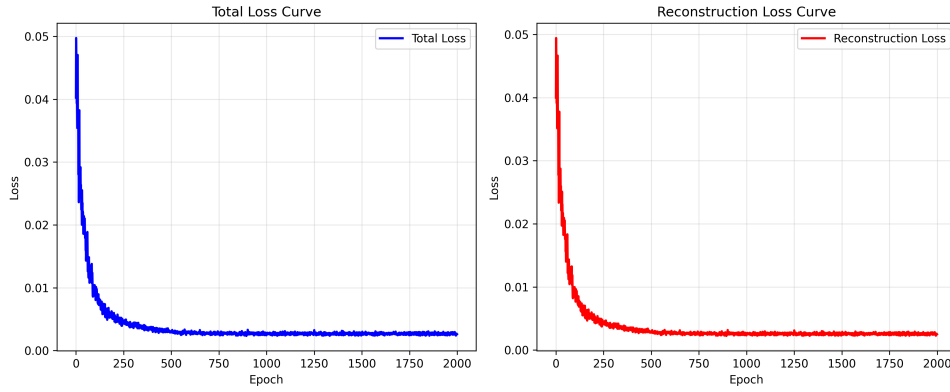


Figure 2: IDRП training loss curves

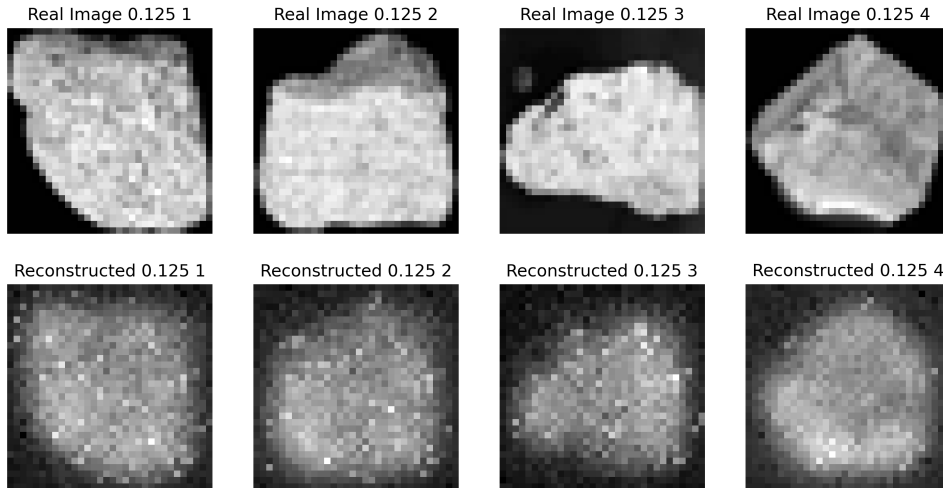


Figure 3: IDRП reconstruction examples

Image 1: PSNR = 15.40 dB, MAE = 0.128622, MSE = 0.028872
 Image 2: PSNR = 14.94 dB, MAE = 0.135155, MSE = 0.032035
 Image 3: PSNR = 13.54 dB, MAE = 0.154129, MSE = 0.044223
 Image 4: PSNR = 17.87 dB, MAE = 0.100705, MSE = 0.016349

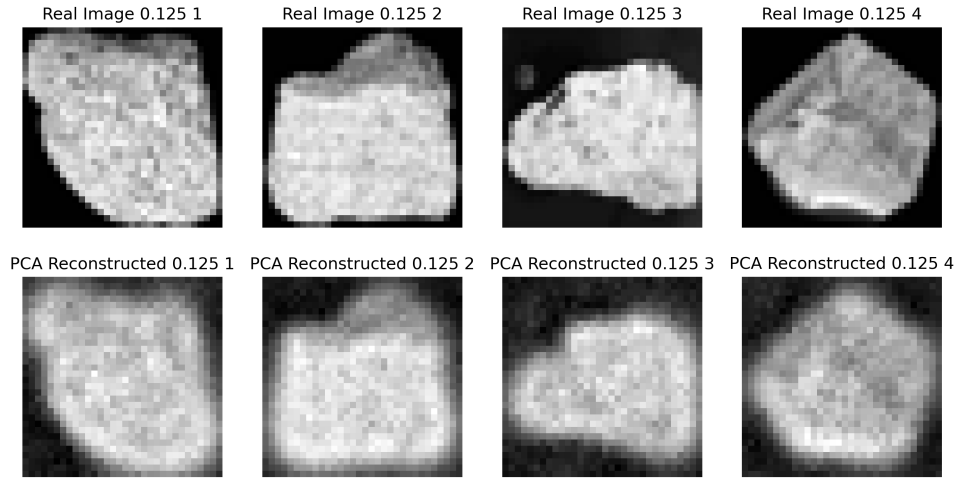


Figure 4: PCA reconstruction examples

4.1.2 PCA:

Image 1: PSNR = 27.68 dB, MAE = 0.032490, MSE = 0.001706

Image 2: PSNR = 27.22 dB, MAE = 0.034426, MSE = 0.001898

Image 3: PSNR = 29.60 dB, MAE = 0.026074, MSE = 0.001097

Image 4: PSNR = 27.90 dB, MAE = 0.031142, MSE = 0.001624

4.2 Real Data Training (PCA for fitting), Real Data Testing

4.2.1 IDR P (Predictor pre-trained for 2000 epochs, 6 layers, 256 hidden dimensions):

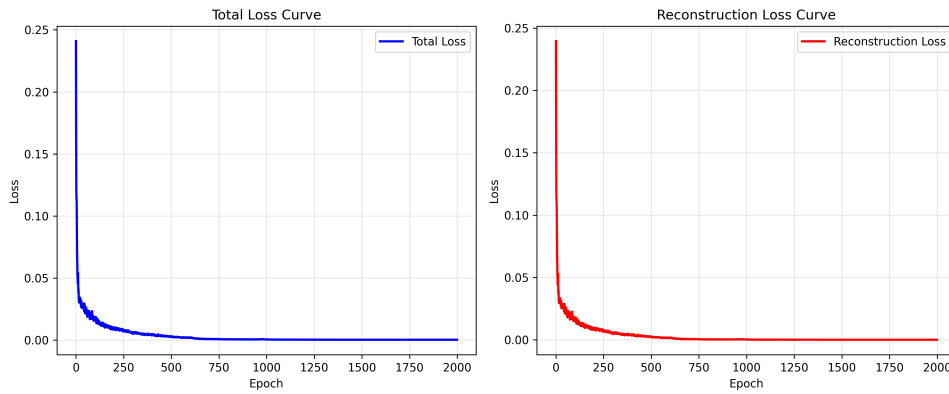


Figure 5: IDR P training loss curves (real data)

Image 1: PSNR = 40.60 dB, MAE = 0.004741, MSE = 0.000087

Image 2: PSNR = 39.04 dB, MAE = 0.005617, MSE = 0.000125

Image 3: PSNR = 44.34 dB, MAE = 0.002972, MSE = 0.000037

Image 4: PSNR = 42.05 dB, MAE = 0.004154, MSE = 0.000062

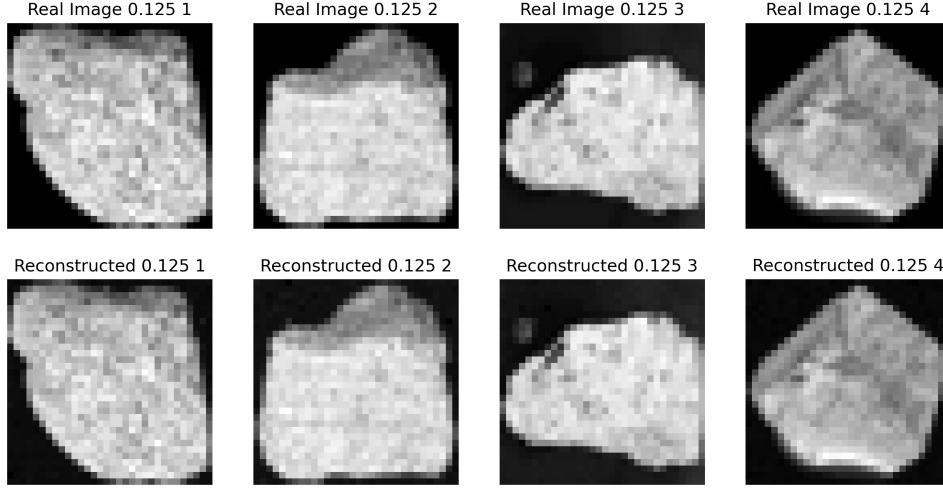


Figure 6: IDR reconstruction examples (real data)

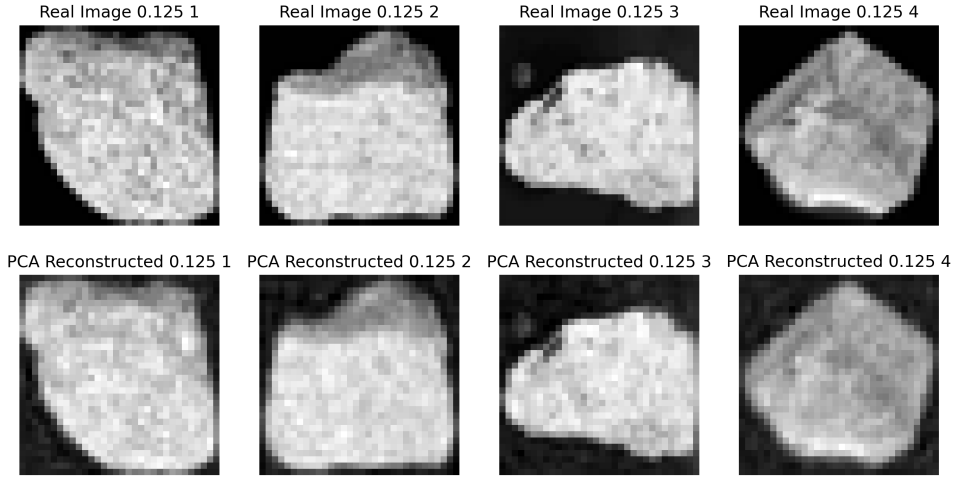


Figure 7: PCA reconstruction examples (real data)

4.2.2 PCA:

Image 1: PSNR = 27.68 dB, MAE = 0.032490, MSE = 0.001706

Image 2: PSNR = 27.22 dB, MAE = 0.034426, MSE = 0.001898

Image 3: PSNR = 29.60 dB, MAE = 0.026074, MSE = 0.001097

Image 4: PSNR = 27.90 dB, MAE = 0.031142, MSE = 0.001624

Experimental Setup and Findings:

- **Baseline:** PCA
- **Metrics:** MSE, PSNR, MAE
- **Key Conclusions:**
 - On nonlinear data, ApCM Model's reconstruction error is significantly lower than PCA's, validating its nonlinear modeling advantage;
 - It shows strong fitting capability on training data, indicating the predictor network possesses good information completion ability.

4.3 Comparison with random data generated by the Key-Value Memory Network

To further evaluate the efficiency and reconstruction performance of ApCM Model, we compare it against a traditional Key-Value Memory Network under random data conditions. The comparison results are summarized in Table 1.

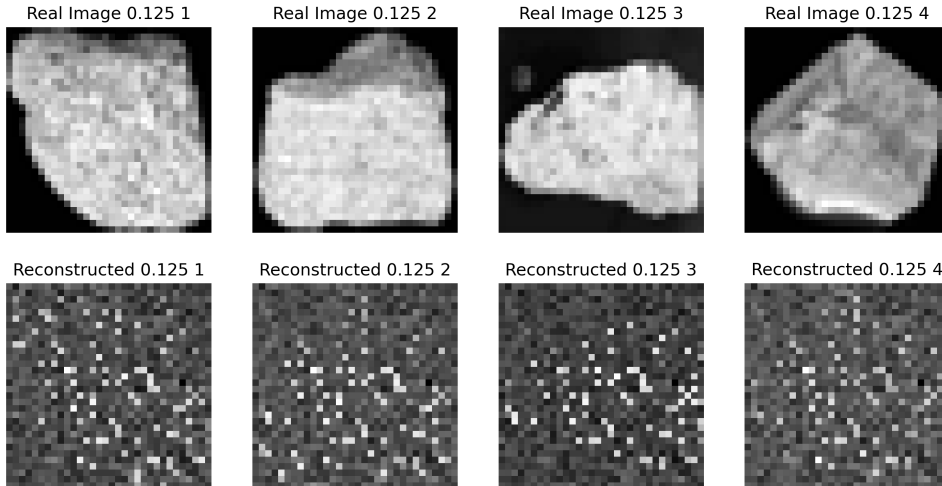
Table 1: Comparison between ApCM Model and Key-Value Memory Network

Metric	ApCM Model	Key-Value Memory
Storage Dimension	128	1024
Number of Memory Slots	32	32
Theoretical Storage Efficiency	High (compressed storage)	Low (full storage)
Test MSE Error	0.987171	1.001440
Test MAE Error	0.765991	0.798507
Inference Time (seconds)	0.1800	0.0010

As shown in the table, ApCM Model achieves comparable reconstruction accuracy (lower MSE and MAE) while maintaining significantly higher storage efficiency due to compressed representation. However, the inference time of ApCM Model is higher than that of the Key-Value Memory baseline, indicating a trade-off between compression efficiency and retrieval speed. This suggests that further optimization of the predictor and retrieval mechanism is needed for real-time applications.

4.4 Ablation Experiment

Reconstructing data directly without predicting z_{aux} is essentially no different from generating data randomly.



5 Reconstruction Noise Analysis

Prediction errors for $\mathbf{z}_{aux}$ during reconstruction propagate to the output, forming **structural noise**, whose characteristics relate to the predictor’s capability boundaries. When inputs deviate from the training distribution, reconstruction quality may degrade. Future

work could explore more robust predictor architectures or introduce uncertainty quantification mechanisms. Experiments indicate that there exists an optimal range for network depth and layer count concerning noise fitting.

6 Conclusion and Future Work

This paper proposes ApCM Model, a neural memory storage model based on invertible compression and learnable prediction, aiming to build an efficient and learnable runtime memory module for AI systems. The model achieves lossless encoding via invertible networks, enables lossy reconstruction with a lightweight predictor, and implements dynamic storage and retrieval with a memory bank. Experiments show that ApCM Model outperforms traditional linear compression methods on nonlinear data, demonstrating stronger modeling capability and stable reconstruction performance.

Future work can proceed in the following directions:

1. **System Integration:** Embed ApCM Model into LLM inference pipelines, exploring its application in tasks like long-context modeling and personalized dialogue.
2. **Predictor Optimization:** Design more robust and efficient auxiliary information prediction networks to improve reconstruction quality and generalization.
3. **Memory Control Learning:** Investigate mechanisms for models to autonomously control memory writing, achieving more human-like memory management and knowledge integration.
4. **Extended Application Scenarios:** Attempt to validate and extend the framework’s utility in tasks such as continual learning and multimodal memory.

ApCM Model provides a new approach to runtime memory modeling. Its learnable and scalable characteristics are expected to advance AI systems with memory capabilities to a higher level.

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References

- [1] Jordan, M. I., & Jacobs, R. A. (1994). Hierarchical mixtures of experts and the EM algorithm. *Neural computation*, 6(2), 181–214.
- [2] Dinh, L., Sohl-Dickstein, J., & Bengio, S. (2017). Density estimation using Real NVP. *ICLR*.

- [3] Papamakarios, G., Nalisnick, E., Rezende, D. J., Mohamed, S., & Lakshminarayanan, B. (2021). Normalizing Flows for Probabilistic Modeling and Inference. *JMLR*.
- [4] Shazeer, N. (2020). GLU Variants Improve Transformer. *arXiv preprint arXiv:2002.05202*.
- [5] Graves, A., Wayne, G., & Danihelka, I. (2014). Neural Turing Machines. *arXiv preprint arXiv:1410.5401*.